A CUBIC HEAT-ENERGY PROOF OF THE MAXIMAL CARLESON ESTIMATE ON L^3

Abstract. We give a detailed heat-energy proof of the maximal modulation estimate for the Hilbert transform on $L^3(\mathbb{R})$. The proof uses the Hessian of the positive trace functional $\operatorname{tr}(U^{3/2})$, a cubic trace inequality, and two correction energies which cancel a mixed-Hessian term. An odd Gaussian factorization supplies the Hilbert kernel. The measurable modulation enters only through fixed unimodular factors, so it is never differentiated. The argument applies the energy method of the bilinear Gaussian notes directly to the linearized form; no boundedness theorem for the general triangular Hilbert transform or time-frequency decomposition is used. The resulting proof still has a three-kernel auxiliary representation. It does not establish the missing adjoint L^3 endpoint or the direct L^2 energy argument.

1. Statement and linearization

Use the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-ix\xi}\, dx, \qquad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(\xi) e^{ix\xi}\, d\xi.$$

For Schwartz functions put

$$H_\alpha f(x) = \frac{1}{\pi}\, \mathrm{pv} \int_{\mathbb{R}} e^{i\alpha t} f(x-t)\, \frac{dt}{t}, \qquad \mathcal{C} f(x) = \sup_{\alpha\in\mathbb{R}} |H_\alpha f(x)|.$$

The multiplier of H_α is $-i\,\mathrm{sgn}(\xi-\alpha)$.

Theorem 1.1. *There is an absolute constant C such that*

$$\|\mathcal{C} f\|_{L^3(\mathbb{R})} \le C \|f\|_{L^3(\mathbb{R})}.$$

The supremum is first defined on Schwartz functions and extends by this estimate to L^3. In particular the maximal Fourier projections onto half-lines and the maximal symmetric Fourier integrals are bounded on L^3. The same estimate holds for the maximal symmetric partial sums of Fourier series on $\mathbb{R}/(2\pi\mathbb{Z})$.

It is enough to prove, uniformly in measurable real N,

$$\left| \int_{\mathbb{R}} H_{N(x)} f(x) u(x) v(x)\, dx \right| \le C \|f\|_3 \|u\|_3 \|v\|_3. \tag{1.1}$$

Indeed any $g \in L^{3/2}$ factors as $g = uv$, with

$$u = |g|^{1/2}, \qquad v = \mathrm{ph}(g) |g|^{1/2}, \qquad \|u\|_3 \|v\|_3 = \|g\|_{3/2}.$$

Conjugations can be included in v. For a finite set of frequencies choose, measurably, the first frequency attaining the largest modulus. Thus a bound uniform in N gives the finite maximal bound. The passage to all frequencies and to general L^3 functions is proved in Section 6.

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2. The Gaussian representation of the odd kernel

Throughout the proof

$$\rho_s(t) = (2\pi s)^{-1/2} e^{-t^2/(2s)}$$

has variance s, so $\partial_s \rho_s = \frac{1}{2}\rho_s''$. Put

$$\psi_s = \sqrt{s}\,\rho_{s/3}', \qquad K_s = s^{3/2}\rho_s'''. \tag{2.1}$$

Gaussian convolution gives $\psi_s * \psi_s * \psi_s = K_s$. Moreover

$$\widehat{K}_s(\xi) = (i\xi)^3 s^{3/2} e^{-s\xi^2/2},$$

and, for $\xi \neq 0$,

$$\begin{aligned}\int_0^\infty \widehat{K}_s(\xi)\,\frac{ds}{s} &= -i\xi^3 \int_0^\infty s^{1/2} e^{-s\xi^2/2}\,ds \\ &= -i\sqrt{2\pi}\,\operatorname{sgn}\xi.\end{aligned} \tag{2.2}$$

For $0 < a < b$ define the bounded integral operator

$$H_N^{[a,b]} f(x) = \frac{1}{\sqrt{2\pi}} \int_a^b \int_{\mathbb{R}} e^{iN(x)t} f(x-t) K_s(t)\,dt\,\frac{ds}{s}. \tag{2.3}$$

At fixed a, b its absolute kernel has L^1 norm at most $C\log(b/a)$; this fact will justify a cutoff limit, not the uniform estimate.

3. The trace Hessian and the cubic inequality

All operators below act on $L^2(\mathbb{R})$ or on a finite direct sum of copies of this space. Write $\|\cdot\|_{\mathrm{HS}}$ for the Hilbert–Schmidt norm and $\|\cdot\|_{\mathrm{op}}$ for the operator norm. For positive trace-class U set

$$\Phi(U) = \operatorname{tr}(U^{3/2}).$$

Its Hessian, on selfadjoint directions, is the possibly extended quadratic form

$$\mathfrak{b}_U(K) = \frac{3}{2}\int_0^\infty \|e^{-t\sqrt{U}/2} K e^{-t\sqrt{U}/2}\|_{\mathrm{HS}}^2\,dt. \tag{3.1}$$

In a spectral basis this reads

$$\mathfrak{b}_U(K) = \frac{3}{2}\sum_{i,j} \frac{|K_{ij}|^2}{\sqrt{\lambda_i} + \sqrt{\lambda_j}}.$$

The coefficient at $i = j$ is $3/(4\sqrt{\lambda_i})$. The form at zero spectrum is interpreted by decreasing positive regularization. Its polarization is denoted by $\mathfrak{b}_U(K, L)$.

We need three properties:

$$U \le V \implies \mathfrak{b}_V(K) \le \mathfrak{b}_U(K), \tag{3.2}$$

$$\mathfrak{b}_{\int U_d\,d\nu}\left(\int K_d\,d\nu\right) \le \int \mathfrak{b}_{U_d}(K_d)\,d\nu, \tag{3.3}$$

$$\mathfrak{b}_{\lambda U}(\lambda K) = \lambda^{3/2}\mathfrak{b}_U(K), \qquad \lambda > 0. \tag{3.4}$$

Here ν is a probability measure. To verify the first two properties, use

$$\mathfrak{b}_U(K) = \frac{3}{2}\sup_{Z = Z^* \in \mathcal{S}^2} \{2\operatorname{tr}(KZ) - 2\operatorname{tr}(\sqrt{U}\,Z^2)\}. \tag{3.5}$$

This is the inverse quadratic form of $Z \mapsto \sqrt{U}Z + Z\sqrt{U}$. The square root is operator increasing, by inverse order and the resolvent integral

$$\sqrt{U} = \frac{1}{\pi}\int_0^\infty \lambda^{-1/2} U(U + \lambda I)^{-1}\,d\lambda.$$

It is operator concave: if $B = \int \sqrt{U_d}\, d\nu$, then

$$\int U_d\, d\nu - B^2 = \int (\sqrt{U_d} - B)^2\, d\nu \geq 0,$$

and square-root monotonicity gives $B \leq \sqrt{\int U_d\, d\nu}$. Inserting these facts into (3.5) proves (3.2) and (3.3). Formula (3.4) follows directly from (3.1).

Lemma 3.1 (Cubic trace inequality)**.** *If $M = M^*$ is Hilbert–Schmidt and $D = D^*$ is bounded, then*

$$|\operatorname{tr}((DM)^3)| \leq 2\|D\|_{\mathrm{op}} \mathfrak{b}_{M^2}(MDM). \tag{3.6}$$

Proof. Put $Q = |M|$, $S = \operatorname{sgn} M$, $E_t = e^{-tQ}$, and $F(t) = \operatorname{tr}((DME_t)^3)$. Differentiation and cyclicity give

$$|F'(t)| = 3|\operatorname{tr}(QME_tDME_tDME_tD)|.$$

For $A_t = ME_tDQE_t^{1/2}$, direct cyclic rearrangement gives

$$\operatorname{tr}(QME_tDME_tDME_tD) = \operatorname{tr}(A_tSA_t^*D).$$

Therefore

$$\begin{aligned} |F'(t)| &\leq 3\|D\|_{\mathrm{op}} \|ME_tDQE_t^{1/2}\|_{\mathrm{HS}}^2 \\ &\leq 3\|D\|_{\mathrm{op}} \|QE_t^{1/2}DQE_t^{1/2}\|_{\mathrm{HS}}^2, \end{aligned}$$

where the last step uses $M^2E_t^2 = Q^2E_t^2 \leq Q^2E_t$ inside a positive trace. Also $F(t) \to 0$: $\|ME_t\|_{\mathrm{op}} \leq 1/(et)$ and $\|ME_t\|_{\mathrm{HS}} \leq \|M\|_{\mathrm{HS}}$. Finally

$$\int_0^\infty \|QE_t^{1/2}DQE_t^{1/2}\|_{\mathrm{HS}}^2\, dt = \frac{2}{3}\mathfrak{b}_{M^2}(MDM).$$

The equality follows by left and right multiplication by S, which preserves the relevant Hilbert–Schmidt norm on the support of Q. Integrating proves the lemma, including the case of an infinite right side. □

For three Hilbert–Schmidt kernels W_v from the $(v+1)$st space to the vth space, indices modulo three, and bounded real multipliers η_v, put

$$\tau = \operatorname{tr}([\eta_0]W_0[\eta_1]W_1[\eta_2]W_2), \qquad R_v = [W_v, W_{v-1}^*].$$

Then

$$|\tau| \leq \frac{\max_v \|\eta_v\|_\infty}{\sqrt{3}} \sum_{v=0}^{2} \mathfrak{b}_{R_v^*R_v}(R_v^*[\eta_v]R_v). \tag{3.7}$$

Here $[\eta]$ means multiplication by η. To prove this, first rotate W_0 by a constant phase ζ so that τ is nonnegative real. The affected rows become

$$R_0 \operatorname{diag}(\zeta I, I), \qquad R_1 \operatorname{diag}(I, \overline{\zeta} I),$$

while R_2 is unchanged. Each Gram and direction is conjugated by the same unitary, so its Hessian expression is unchanged. The rotation is used only in this algebraic inequality at the fixed parameters: no rotated kernel is differentiated later. Form the selfadjoint block matrix M with off-diagonal blocks $M_{v,v+1} = W_v$ and their adjoints, and put $D = \operatorname{diag}([\eta_v])$. Then

$$\operatorname{tr}((DM)^3) = 6|\tau|, \quad M^2 = \sum_v MP_vM, \quad MDM = \sum_v MP_vDM,$$

where P_v projects onto the vth space. By (3.3) and (3.4),

$$\mathfrak{b}_{M^2}(MDM) \leq \sqrt{3} \sum_v \mathfrak{b}_{MP_vM}(MP_vDM).$$

The summands are the row expressions in (3.7). Apply Lemma 3.1 and divide by six.

4. The corrected heat energy

Fix $u(x,y), v(x,z) \in L^3(\mathbb{R}^2)$. At scale s and centers $p = (p_0, p_1, p_2)$ define

$$C(x,y) = u(x,y)\sqrt{\rho_{s/6}(x-p_0)\rho_s(y-p_1)},$$
$$D(x,z) = v(x,z)\sqrt{\rho_{s/6}(x-p_0)\rho_s(z-p_2)}.$$

Set

$$R = [C, D], \qquad A = CC^*, \quad B = DD^*, \quad T = R^*R, \qquad w_s(p) = \rho_{5s/6}(p_0+p_1+p_2).$$

The two Grams T and $A+B = RR^*$ have the same nonzero spectrum. Define

$$E(s) = \int_{\mathbb{R}^3} [\Phi(A+B) + \tfrac{1}{2}\Phi(A) + \tfrac{1}{2}\Phi(B)] w_s(p)\, dp. \tag{4.1}$$

Proposition 4.1. *With $c_E = \frac{1}{2} + \sqrt{2}$, one has*

$$0 \le E(s) \le c_E(\|u\|_3^3 + \|v\|_3^3), \tag{4.2}$$

$$E'(s) + \frac{1}{12}\int_{\mathbb{R}^3} \mathfrak{b}_T(\partial_{p_0} T) w_s\, dp \le 0. \tag{4.3}$$

The differential statement is understood in its integrated form: for every $0 < a < b$,

$$\frac{1}{12}\int_a^b \int_{\mathbb{R}^3} \mathfrak{b}_{T_s}(\partial_{p_0} T_s) w_s\, dp\, ds \le E(a) - E(b).$$

The proof below performs the calculation with a regularized trace and then passes to this inequality.

Proof. Schatten Holder and scalar convexity give

$$\Phi(A+B) \le \sqrt{2}[\Phi(A) + \Phi(B)], \qquad \Phi(CC^*) \le \|C\|_{\mathrm{HS}}^3.$$

Since $\rho_{s/6}(x-p_0)\rho_s(y-p_1)\,dx\,dy$ is a probability measure, Jensen gives

$$\|C\|_{\mathrm{HS}}^3 \le \iint |u(x,y)|^3 \rho_{s/6}(x-p_0)\rho_s(y-p_1)\,dx\,dy.$$

Integrating the unused center p_2 out of w_s, and then integrating p_0, p_1, gives $\int \Phi(A) w_s \le \|u\|_3^3$. The analogous bound for B proves (4.2).

We give the derivative calculation explicitly. If U depends linearly on a Gaussian density of variance σ and center p, then $\partial_\sigma U = \frac{1}{2}\partial_p^2 U$, and the trace chain rule yields

$$(\partial_\sigma - \tfrac{1}{2}\partial_p^2)\Phi(U) = -\tfrac{1}{2}\mathfrak{b}_U(\partial_p U). \tag{4.4}$$

For the common variable use T, which is linear in the common density. For the two outer variables use $A+B$, which is linear in each corresponding density. This is legitimate because $\Phi(T) = \Phi(A+B)$ for every independent choice of the three variances.

Write $F = \Phi(T) = \Phi(A+B)$, $A_1 = \partial_{p_1} A$, $B_2 = \partial_{p_2} B$, and $Q = \mathfrak{b}_T(\partial_{p_0} T)$. Along the variances $(s/6, s, s)$, (4.4) gives

$$\begin{aligned}\partial_s F = {} & \tfrac{1}{12}\partial_{p_0}^2 F + \tfrac{1}{2}\partial_{p_1}^2 F + \tfrac{1}{2}\partial_{p_2}^2 F \\ & - \tfrac{1}{12} Q - \tfrac{1}{2}\mathfrak{b}_{A+B}(A_1) - \tfrac{1}{2}\mathfrak{b}_{A+B}(B_2).\end{aligned} \tag{4.5}$$

Also $\partial_s w_s = \frac{5}{12} w_s''$, where the double prime differentiates its scalar argument. Integrating by parts, the four Gaussian derivative coefficients add to

$$\tfrac{1}{12} + \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{5}{12} = \tfrac{3}{2}.$$

Since $\partial_{p_1}\partial_{p_2} w_s = w_s''$ and A is independent of p_2 while B is independent of p_1,

$$\int F w_s''\, dp = \int \partial_{p_1}\partial_{p_2} F\, w_s\, dp = \int \mathfrak{b}_{A+B}(A_1, B_2) w_s\, dp.$$

Consequently

$$\frac{d}{ds}\int F w_s\, dp + \frac{1}{12}\int Q w_s\, dp = \int \big[\tfrac{3}{2}\mathfrak{b}_{A+B}(A_1, B_2) - \tfrac{1}{2}\mathfrak{b}_{A+B}(A_1) - \tfrac{1}{2}\mathfrak{b}_{A+B}(B_2)\big] w_s\, dp. \tag{4.6}$$

Cauchy–Schwarz for the nonnegative Hessian gives

$$2\mathfrak{b}_{A+B}(A_1, B_2) \le \mathfrak{b}_{A+B}(A_1) + \mathfrak{b}_{A+B}(B_2).$$

Together with (3.2), the right side of (4.6) is at most

$$\frac{1}{4}\int [\mathfrak{b}_A(A_1) + \mathfrak{b}_B(B_2)] w_s\, dp. \tag{4.7}$$

For the separate A energy first integrate out p_2. The remaining center integrals are unweighted. Apply (4.4), using C^*C for the common variable and CC^* for the outer variable. Their second derivatives integrate to zero, leaving

$$\frac{d}{ds}\int \Phi(A) w_s\, dp \le -\frac{1}{2}\int \mathfrak{b}_A(A_1) w_s\, dp.$$

Similarly the separate B energy contributes at most $-\frac{1}{2}\int \mathfrak{b}_B(B_2) w_s$. Their coefficients $1/2$ in (4.1) cancel (4.7) exactly, proving (4.3). The justification of differentiation at singular spectra and of integration by parts is supplied in Section 7. □

5. Odd Gaussian pairing controlled by the energy

For fixed kernels $F_0(x_0, x_1), F_1(x_1, x_2), F_2(x_2, x_0)$ in $L^3(\mathbb{R}^2)$ put

$$J_s = \iiint F_0(x_0, x_1) F_1(x_1, x_2) F_2(x_2, x_0) K_s(x_0 + x_1 + x_2)\, dx_0\, dx_1\, dx_2.$$

These integrals are absolutely convergent. In fact, for $k \in L^1$, fixing $t = x_0 + x_1 + x_2$ and applying Holder in the other two variables gives

$$\iiint |F_0 F_1 F_2 k(x_0 + x_1 + x_2)|\, dx \le \|k\|_1 \prod_{v=0}^{2} \|F_v\|_3. \tag{5.1}$$

All three changes of variables from the fixed-t plane to an edge pair have determinant of absolute value one.

At centers $p_0 + p_1 + p_2 = 0$, with measure $dp_0 dp_1$, set

$$q_v(x) = \rho_s(x - p_v), \quad W_v(x_v, x_{v+1}) = F_v(x_v, x_{v+1})\sqrt{q_v(x_v) q_{v+1}(x_{v+1})},$$

and

$$\eta_v(x) = \frac{\psi_s(x - p_v)}{\rho_s(x - p_v)}.$$

Writing $z = (x - p_v)/\sqrt{s}$, direct differentiation gives

$$\eta_v(x) = -3\sqrt{3}\, z e^{-z^2}, \qquad \|\eta_v\|_\infty = 3\sqrt{\frac{3}{2e}} < 3.$$

The convolution identity (2.1) implies

$$J_s = \int_{p_0+p_1+p_2=0} \operatorname{tr}([\eta_0] W_0 [\eta_1] W_1 [\eta_2] W_2)\, dp_0 dp_1. \tag{5.2}$$

Fubini follows from (5.1) applied to $|\psi_s| * |\psi_s| * |\psi_s|$.

Fix a row index v. Let $T^{\text{base}} = R_v^* R_v$ and $L^{\text{base}} = R_v^*[\eta_v] R_v$ for these variance-s kernels. Replace the shared Gaussian variance by $s/6$ and translate its center by d, leaving the outer variances unchanged; denote the resulting correlation by T_d. Gaussian convolution gives

$$T^{\text{base}} = \int T_d \rho_{5s/6}(d)\, dd, \qquad L^{\text{base}} = -\sqrt{s} \int \partial_{p_v} T_d \rho_{s/6}(d)\, dd. \tag{5.3}$$

For the second identity, differentiation in a Gaussian center changes the sign, and $\rho_{s/6} * \rho_{s/6} = \rho_{s/3}$. Also

$$\frac{\rho_{s/6}(d)}{\rho_{5s/6}(d)} \le \sqrt{5}.$$

Apply (3.3) with probability density $\rho_{5s/6}$ and direction $-\sqrt{s}\,[\rho_{s/6}/\rho_{5s/6}]\partial_{p_v} T_d$. Quadratic dependence on the direction gives

$$\mathfrak{b}_{T^{\text{base}}}(L^{\text{base}}) \le 5s \int \mathfrak{b}_{T_d}(\partial_{p_v} T_d) \rho_{5s/6}(d)\, dd.$$

Changing from the two plane centers and d to three unrestricted centers has Jacobian of absolute value one. The density becomes $w_s(p)$, since the new center sum is d. Thus

$$\int_{p_0+p_1+p_2=0} \mathfrak{b}_{T^{\text{base}}}(L^{\text{base}})\, dp_0 dp_1 \le 5s \int_{\mathbb{R}^3} Q_{v,s}(p) w_s(p)\, dp, \tag{5.4}$$

where $Q_{v,s} = \mathfrak{b}_{T_{v,s}}(\partial_{p_v} T_{v,s})$ is the curvature of the refined row in Proposition 4.1.

Combining (3.7), (5.2), and (5.4), then integrating Proposition 4.1, gives

$$\begin{aligned} \int_a^b |J_s| \frac{ds}{s} &\le 5\sqrt{3} \sum_{v=0}^{2} \int_a^b \int Q_{v,s} w_s\, dp\, ds \\ &\le 60\sqrt{3} \sum_{v=0}^{2} E_v(a). \end{aligned} \tag{5.5}$$

Normalize the three input norms to one before applying this inequality. Each of the two kernels in a row then has norm one, so $\sum_v E_v(a) \le 6c_E$. Trilinear homogeneity yields

$$\boxed{\int_a^b |J_s| \frac{ds}{s} \le C_0 \prod_{v=0}^{2} \|F_v\|_3, \qquad C_0 = 360\sqrt{3}(\tfrac{1}{2} + \sqrt{2}).} \tag{5.6}$$

The constant is not optimized. This estimate has been derived from the heat energy, not invoked as a triangular-transform theorem. The only application needed next is to the specified modulation-dependent kernels.

6. Insertion of the modulation and removal of the cutoffs

Fix a measurable N, and let $\chi_R = \mathbf{1}_{[-R,R]}$. For $f, u, v \in L^3(\mathbb{R})$ take

$$\begin{aligned} F_0(x,y) &= u(x) e^{iN(x)(x+y)} \chi_R(y), \\ F_1(y,z) &= f(-y-z) \chi_R(y), \\ F_2(z,x) &= v(x) e^{iN(x)z} \chi_R(z). \end{aligned} \tag{6.1}$$

They are fixed functions of their variables throughout the heat evolution. Their norms satisfy

$$\prod_{v=0}^{2} \|F_v\|_3 = 2R \|f\|_3 \|u\|_3 \|v\|_3. \tag{6.2}$$

In the product of their phases the exponent is $iN(x)(x+y+z)$. Set $t = x+y+z$; then $f(-y-z) = f(x-t)$ and

$$\int \chi_R(y)^2 \chi_R(t-x-y)\,dy = (2R - |t-x|)_+.$$

Consequently, with $f_R(w) = (1-|w|/(2R))_+ f(w)$, the exact identity is

$$\frac{J_s}{2R} = \int u(x)v(x) \int e^{iN(x)t} f_R(x-t) K_s(t)\,dt\,dx. \tag{6.3}$$

Apply (5.6) and (6.2), and divide by $2R\sqrt{2\pi}$. This gives

$$\left| \int H_N^{[a,b]} f_R\, uv \right| \le \frac{C_0}{\sqrt{2\pi}} \|f\|_3 \|u\|_3 \|v\|_3.$$

For fixed a, b, the absolute integrable kernel in (2.3) gives L^3 continuity uniformly in N, with a bound allowed to depend on a, b. As $f_R \to f$ in L^3, let $R \to \infty$. Factoring an arbitrary dual function as in Section 1 proves

$$\|H_N^{[a,b]} f\|_3 \le \frac{C_0}{\sqrt{2\pi}} \|f\|_3 \tag{6.4}$$

uniformly in a, b and N.

For a constant modulation α, the multiplier of (2.3) is

$$-i\,\mathrm{sgn}(\xi-\alpha)\,\frac{1}{\Gamma(3/2)} \int_{a(\xi-\alpha)^2/2}^{b(\xi-\alpha)^2/2} r^{1/2} e^{-r}\,dr.$$

The scalar factor is between zero and one and tends to one away from $\xi = \alpha$ as $a \downarrow 0$, $b \uparrow \infty$. For Schwartz f, dominated convergence against $|\widehat{f}|$ gives pointwise convergence for every fixed pair (x, α). Hence it also gives convergence with $\alpha = N(x)$, without any regularity assumption on N. Fatou in (6.4) yields

$$\|H_{N(\cdot)} f\|_3 \le \frac{C_0}{\sqrt{2\pi}} \|f\|_3.$$

Choose a maximizing selector for any finite frequency set. Enumerate the rational frequencies and use monotone convergence. For Schwartz f, the formula

$$H_\alpha f(x) = \frac{1}{2\pi} \int -i\,\mathrm{sgn}(\xi-\alpha) \widehat{f}(\xi) e^{ix\xi}\,d\xi$$

is continuous in α, so the rational and real suprema agree. This proves Theorem 1.1 for Schwartz functions.

To extend the estimate, approximate $f \in L^3$ by Schwartz f_n. The inequality

$$|\mathcal{C} f_n - \mathcal{C} f_m| \le \mathcal{C}(f_n - f_m)$$

shows that the maximal functions are Cauchy in L^3. Each fixed rational-modulation Hilbert transform also converges in L^3, by its ordinary scalar bound. The finite-frequency maximal inequalities pass to the limit and then, by monotone convergence, bound the rational supremum by the same constant.

There is a direct common-nullset realization for all real modulations. For $\delta \neq 0$, the function $(e^{i\delta t} - 1)/t$ belongs to $L^{3/2}(\mathbb{R})$ and has norm $C|\delta|^{1/3}$. Thus Holder makes the difference integral

$$H_\alpha f(x) - H_\beta f(x) = \frac{1}{\pi} \int \frac{e^{i\alpha t} - e^{i\beta t}}{t}\, f(x-t)\,dt$$

absolutely convergent, with

$$\|H_\alpha f - H_\beta f\|_\infty \le C|\alpha - \beta|^{1/3} \|f\|_3.$$

Choose one representative of $H_0 f$ and define all the others by these differences. This agrees almost everywhere with the ordinary modulated Hilbert transform for each fixed α, and is continuous in α

outside the single nullset for $H_0 f$. The rational and real suprema therefore agree. This completes the L^3 extension without using a maximal Carleson bound in its construction.

For clarity, if $P_{(-\infty,\alpha)}$ is the Fourier projection onto a half-line, then

$$P_{(-\infty,\alpha)} = \tfrac{1}{2}(I - iH_\alpha), \qquad \sup_\alpha |P_{(-\infty,\alpha)} f| \le \tfrac{1}{2}(|f| + \mathcal{C}f).$$

Conversely $H_\alpha = -i(I - 2P_{(-\infty,\alpha)})$. The symmetric Fourier integral is

$$\frac{1}{2\pi}\int_{-R}^{R} \widehat{f}(\xi) e^{ix\xi}\, d\xi = P_{(-\infty,R)} f - P_{(-\infty,-R)} f,$$

so its maximal operator is bounded by twice the maximal half-line projection. These identities specify exactly which classical maximal Fourier integral operators are covered.

The Fourier-series statement follows by localization. Choose a nonzero Schwartz function χ with Fourier support in $[-1/4, 1/4]$. Let $p(x) = \sum_k c_k e^{ikx}$ be a trigonometric polynomial. For $L \ge 1$, put $f_L(x) = \chi(x/L)p(x)$. Each summand has Fourier support in the interval $[k - 1/(4L), k + 1/(4L)]$. Write $S_n p = \sum_{|k|\le n} c_k e^{ikx}$. These support inclusions give the exact identity

$$P_{(-n-1/2,n+1/2)} f_L(x) = \chi(x/L) S_n p(x).$$

The real-line maximal estimate therefore gives

$$\int_{\mathbb{R}} |\chi(x/L)|^3 \sup_{n\ge 0} |S_n p(x)|^3\, dx \le C^3 \int_{\mathbb{R}} |\chi(x/L)|^3 |p(x)|^3\, dx.$$

For any continuous 2π-periodic W, partitioning into periods (or uniformly approximating W by trigonometric polynomials) gives

$$\lim_{L\to\infty} \frac{1}{L}\int |\chi(x/L)|^3 W(x)\, dx = \left(\int_{\mathbb{R}} |\chi|^3\right) \frac{1}{2\pi}\int_0^{2\pi} W(x)\, dx.$$

Apply this to the two continuous periodic functions in the preceding inequality; the supremum is finite for a polynomial. Dividing by the positive factor $\int |\chi|^3$ gives

$$\left\| \sup_{n\ge 0} |S_n p| \right\|_{L^3(\mathbb{R}/(2\pi\mathbb{Z}))} \le C \|p\|_{L^3(\mathbb{R}/(2\pi\mathbb{Z}))}.$$

For general L^3 functions, pass first to each finite supremum by polynomial approximation and then to all n by monotone convergence. Each fixed partial sum is a bounded finite-rank operator, so this passage uses no maximal theorem in advance.

7. Justification of the Gaussian trace calculus

We include the analytic justification so that no differentiability of an arbitrary L^3 kernel or of the selector is being assumed. At a fixed positive scale, the weighted kernels C, D are Hilbert–Schmidt by Holder. Their Grams are positive trace class. Trace cyclicity above is applied only to trace-class products.

For $\varepsilon > 0$ define

$$f_\varepsilon(t) = (t+\varepsilon)^{3/2} - \varepsilon^{3/2} - \tfrac{3}{2}\sqrt{\varepsilon}\, t, \qquad \Phi_\varepsilon(U) = \operatorname{tr} f_\varepsilon(U).$$

This is one trace-class expression, $f_\varepsilon(0) = 0$, and $0 \le f_\varepsilon(t) \uparrow t^{3/2}$ as $\varepsilon \downarrow 0$. Thus the equality of trace functions on the two Grams remains valid, including their different nullspaces. The square-root resolvent formula gives

$$D\Phi_\varepsilon(U)[K] = \tfrac{3}{2} \operatorname{tr}[(\sqrt{U+\varepsilon I} - \sqrt{\varepsilon} I)K],$$
$$D^2\Phi_\varepsilon(U)[K,L] = \mathfrak{b}_{U+\varepsilon I}(K,L).$$

The ordinary chain rule therefore proves the regularized heat and mixed-derivative identities.

Here are bounds allowing the passage to the limit and integration by parts. Write

$$U = \int K_z \rho_\sigma(z-p)\,dz, \qquad K_z \geq 0, \quad a = (z-p)/\sqrt{\sigma}.$$

Trace convexity and (3.3) give

$$\Phi_\varepsilon(U) \leq \int \Phi(K_z)\rho_\sigma(z-p)\,dz,$$
$$\mathfrak{b}_{U+\varepsilon I}(\partial_p U) \leq \frac{3}{4\sigma}\int a^2 \Phi(K_z)\rho_\sigma(z-p)\,dz.$$

For the second line, average the base operators $K_z + \varepsilon I$ and the directions $aK_z/\sqrt{\sigma}$, and use $\mathfrak{b}_{K+\varepsilon I}(K) \leq \frac{3}{4}\Phi(K)$. The variational formula applies equally to these bounded positive regularizations. For positive K, Schatten Holder and Young also imply

$$|D\Phi_\varepsilon(U)[K]| \leq \tfrac{1}{2}\Phi(U) + \Phi(K).$$

Insert the first and second Gaussian derivatives into this estimate and absorb polynomial factors into a Gaussian of twice the variance. Uniformly in ε,

$$|\partial_p \Phi_\varepsilon(U)| \lesssim \sigma^{-1/2}\int \Phi(K_z)\rho_{2\sigma}(z-p)\,dz,$$
$$|\partial_\sigma \Phi_\varepsilon(U)| + |\partial_p^2 \Phi_\varepsilon(U)| \lesssim \sigma^{-1}\int \Phi(K_z)\rho_{2\sigma}(z-p)\,dz. \tag{7.1}$$

The curvature has the same bound as the second line.

In the common variable the operator K_x is rank one, and

$$\Phi(K_x) \leq \sqrt{2}\left[\int |u(x,y)|^3 \rho_s(y-p_1)\,dy + \int |v(x,z)|^3 \rho_s(z-p_2)\,dz\right].$$

In an outer variable write the Gram as the average of $K_y + B$; K_y is rank one and

$$\Phi(K_y) \leq \int |u(x,y)|^3 \rho_{s/6}(x-p_0)\,dx, \quad \Phi(K_y+B) \leq \sqrt{2}[\Phi(K_y)+\Phi(B)].$$

Each scalar term omits one center. Integrate that center out of the center-sum Gaussian, then integrate the remaining Gaussian densities. The resulting first-derivative bounds are at most $Cs^{-1/2}(\|u\|_3^3 + \|v\|_3^3)$; second derivatives and curvatures have the corresponding Cs^{-1} bound. The same reasoning works when any Gaussian variance is enlarged by a fixed factor and when the center-sum variance is any positive multiple of s.

These estimates justify removal of center cutoffs in integration by parts and differentiation of the regularized center integrals on compact positive scale intervals. It is simplest to keep Φ_ε throughout the calculation in Proposition 4.1. Its mixed term uses $\mathfrak{b}_{A+B+\varepsilon I}$; order bounds it by the two pure curvatures $\mathfrak{b}_{A+\varepsilon I}$ and $\mathfrak{b}_{B+\varepsilon I}$, which are canceled by the separate regularized energies. Thus, writing E_ε for (4.1) with Φ_ε,

$$E_\varepsilon(a) - E_\varepsilon(b) \geq \frac{1}{12}\int_a^b \int \mathfrak{b}_{T_s+\varepsilon I}(\partial_{p_0} T_s) w_s\,dp\,ds.$$

The endpoints increase to $E(a)$ and $E(b)$ individually and are finite by (4.2). The nonnegative quadratic forms increase to $\mathfrak{b}_{T_s}$. Monotone convergence therefore gives the required integrated inequality directly. No unregularized second-derivative assertion is needed for the proof.

8. What the exponent three does in this proof

The identity $\Phi(R^*R) = \|R\|_{\mathcal{S}^3}^3$ matches the cubic trace of three kernels. It also matches the factorization $L^{3/2} = L^3 \cdot L^3$ in (1.1). These are the two uses of the exponent three. The two separate energies in (4.1) supply the mixed-Hessian cancellation; this is the part of the bilinear method used here.

The proof establishes the primal maximal estimate. It does not assert a uniform L^3 estimate for the adjoint of every linearization. Such an adjoint estimate, together with the result proved here, would give the L^2 result by duality and interpolation, but it is an additional endpoint and has not been proved by this calculation.

Provenance. The trace and heat mechanism is expanded from the supplied bilinear Gaussian variation notes (version 4), especially their heat-energy calculation and operator appendix. The odd Gaussian factorization and modulation-dependent kernels are the adaptation recorded in the Carleson energy research checkpoint (version 11). All analytic estimates needed for the L^3 argument have been written out above; no general triangular-transform boundedness result is invoked.